Learning Starts From Optimizing the Composition of Temporal Information for Hyperspectral Change Detection

Yaxiong Chen, Zhipeng Zhang, Jirui Huang, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— Hyperspectral image change detection (HSI-CD) is a task that utilizes both spectral and spatial features to more effectively detect changes. The features of HSI captured at different times are often influenced by external factors. The annoying variability is not useful for CD. Current methods extract effective information through complex learning together with it. They did not focus on whether different categories (changed and unchanged) of sample pairs have the same information composition. They lack a fundamental analysis and optimization based on the composition of information pairs to address current issues such as insufficient recognition of changed samples and poor feature fusion. To address these issues, we rethink the information composition of bitemporal sample pairs in different categories for HSI-CD, and design a frequency-domain information exchange and generation network with Siamese U-shaped structure (FDIEG-UNet) for HSI-CD. Our contribution can be summarized as follows: 1) by designing the FDGJLS module for learning and separating global–joint spectral–spatial features, to provide better basic frequency-domain information for subsequent feature-domain optimization; 2) based on the information from FDGJLS, we use the ULDT and SSTDFE modules to remove the influence of temporal-correlated invalid style information in feature fusion. Two modules work together to improve the equality of feature description between changed and unchanged samples in CD; and 3) the experimental results on three real HSI-CD datasets demonstrate the effectiveness of our proposed method in solving the above problems. The code is available at https://github.com/WUTCM-Lab/FDIEG-UNet.**

***Index Terms*— Change detection, feature domain, frequency domain, hyperspectral image (HSI), information composition.**

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I. Introduction

HYPERSPECTRAL image change detection (HSI-CD) is an important task that utilizes the high spectral and spatial resolution of hyperspectral images (HSIs) taken at different times to detect changes in the same scene. By using hyperspectral sensors mounted on different space platforms, the target area is simultaneously imaged in tens to hundreds of continuous and subdivided spectral bands in the ultraviolet, visible, near-infrared, and mid-infrared regions of the electromagnetic spectrum [1]. In recent years, HSI-CD has been widely studied and applied in various fields such as agricultural development, vegetation monitoring, urban planning, and disaster assessment [2], [3], due to its characteristic of having hundreds of continuous spectral bands from ultraviolet to infrared in HSIs.

The mainstream method of early CD involved extensive image processing and calculation, and the application of appropriate thresholds to detect changes, including algebra [4], transformation [5], and classification [6]. However, using only handcrafted shallow features cannot represent the key information in HSI, resulting in poor detection accuracy. In recent years, with the development of big data, various deep learning networks such as convolutional neural network (CNN) [7], recurrent neural network (RNN) [8], and transformer [9] have been widely used in HSI-CD.

At present, research on HSI-CD is still in its early stages. Numerous methods focus on time-series-related feature fusion optimization and feature extraction optimization related to

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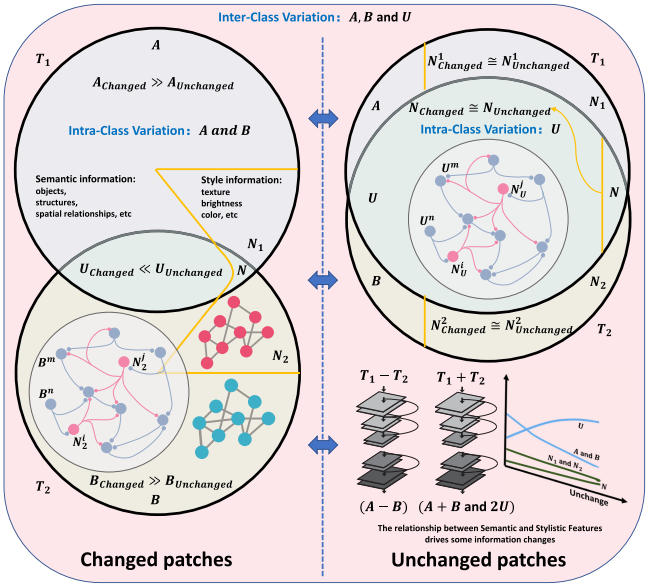


Fig. 1. Explanation of the new perspectives on the information composition in bitemporal HSI patches. **On the left**, it represents the information structure of pairs of changed patches. **On the right**, it represents the information structure of pairs of unchanged patches.

data structure [10]. The most challenging part of HSI-CD is tracking the changes that users want to highlight. The current methods still have the following issues regarding this challenge.

1) *In feature extraction:* global–joint temporal–spectral–spatial information is utilized. Recent studies have shown that combining temporal–spectral–spatial features with HSI structures is beneficial for improving performance [11], [12], [13]. The current methods focus on 3-D CNN or cross-attention to extract joint features [14], [15], [16], [17]. But this only preserves the local information of the joint features. For current patch-based methods, they precisely lack attention to global information. How to combine joint features with the advantages of transformers for global spectral feature extraction is a worthwhile research question.
2) *In feature fusion:* there are often changes in spectral and spatial information caused by external factors (such as environment, lighting conditions, or seasonal changes) [18]. These pieces of information do not differ significantly between changed and unchanged sample pairs, and contribute almost no to inter-class differences. This often leads to poor feature fusion performance. Some methods focus on feature extraction to address this issue, using denoising or redundancy to improve the effectiveness of extracting semantic features [19], [20], [21], [22].
3) *In feature fusion:* the recognition ability of changed samples is much lower than that of unchanged samples. Unchanged pixels are often time-independent, while changing pixels often require time-related feature comparison and fusion. The current methods focus on optimizing the recognition performance of changing samples through methods such as comparing or fusing temporal features [16], [23], [24], [25].

Specifically, Chen et al. [23] achieved information fusion of bitemporal features through bidirectional interaction between transformers. Zhang et al. [16] interacted features using simple concatenation and sequential learning. With further improvements, Zhang et al. [25] captured the correlation between bitemporal HSI features from more perspectives such as difference and dot product operations. Chen et al. [19] combined temporal spatial spectral information through cross-attention. However, the above-mentioned methods simply and crudely extract and fuse the overall information to extract effective information. They did not focus on the basic structure of the information itself. Nor did it consider whether the components of the information made sufficient contributions to CD, or if they were fair enough in the process of identifying changing or unchanged samples. We knew from the beginning that there are some uncertain factors that cannot provide information for the task at hand. This forces the model to learn together with another annoying variability, which fundamentally limits the improvement of CD performance. The known interference variability can be addressed from the beginning without the need for complex training to learn.

Here, we propose a new approach to explain the information composition of bitemporal HSIs, aiming to optimize the equality and saliency of features in the HSI-CD based on this explanation. Our motivation for introducing semantic and style information is based on the spectral absorption and reflection characteristics of different natural scenes in bitemporal HSI.

Image-to-image translation and style transformation are commonly used to improve the domain adaptability of semantic segmentation [26]. Through research in domain adaptation, we can consider that a feature map can be divided into style features and semantic features [27]. Semantic information includes object details, structural information, spatial and spectral position relationships, and so on. Style information includes texture, brightness, color information, and so on. Style information and semantic information are related, and we learn high-level structural information, and so on through the network to obtain effective high-level features. However, there are invalid style information caused by imaging differences that are not related to CD, such as brightness in the style information. We need to remove it.

As shown in Fig. 1, we can consider the spectral and spatial feature domains to be composed of specific information in the patch-based method. The unique information of HSI patch T_1 comprises semantic information A and invalid style information N_1, while that of T_2 comprises semantic information B and invalid style information N_2. The common information between T_1 and T_2 consists of semantic information U and style information N. In the subsequent equations, the subscripts "changed" and "unchanged" denote the U, A, and B in the changed and unchanged samples, respectively.

As shown on the left of Fig. 1, for changed samples, there still exists limited common information in pairs of changed patches. However, concerning inter-class variation between changed and unchanged samples, $U_{\text{Changed}} \ll U_{\text{Unchanged}}$. As shown on the right of Fig. 1, in the case of unchanged samples, limited unique information still exists in pairs of unchanged patches, but $A_{\text{Changed}} \gg A_{\text{Unchanged}}$ and $B_{\text{Changed}} \gg B_{\text{Unchanged}}$. Invalid style information is only related to external

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factors of T_1 and T_2. It can vary significantly without affecting the perception of high-level semantics. Whether an object is a vehicle or a person should not depend on the characteristics of sensors, light sources, or other low-level sources of change, so $N^1_{\text{Changed}} \cong N^1_{\text{Unchanged}}$ and $N^2_{\text{Changed}} \cong N^2_{\text{Unchanged}}$.

As shown in the curve in the bottom right corner of Fig. 1, with the increase of unchanged samples, we observe that the semantic information A, B, and U contribute significantly to the inter-class variation, while the contribution of invalid style information to CD is not significant. Single temporal information is not sufficient, and we need to perform temporal-related feature fusion for CD through the following methods:

$$T_1 + T_2 = f(A + B + 2U, N_1 + N_2) \tag{1}$$

where $f(\cdot)$ is the representation of style information and semantic information in the feature domain. We need to use DL to remove the influence of invalid style information to obtain saliency original information for CD.

Meanwhile, U contributes significantly to the intra-class variation of unchanged samples. A and B contribute more to the intra-class variation in the changed samples. Single U can fully reflect all information in unchanged samples, but for bitemporal information A and B, simple summation of feature fusion may offset some key information, which is not fair for detecting changed samples. Similarly, we need to explore other methods to enhance differential information

$$T_1 - T_2 = f(A - B, N_1 - N_2). \tag{2}$$

Similarly, we need to use DL to remove invalid style information and combine it with the first feature fusion method mentioned above to enhance the equality of CD.

So, we need to separate style features and semantic features in order to achieve simpler and more effective feature fusion. Inspired by the Fourier transform, we use it to separate two types of information. In order to provide more effective features for optimizing feature fusion, we use frequency domain to focus on global–joint temporal–spectral–spatial features in feature extraction. Then, two feature fusion methods that are not affected by annoying variability are achieved through learning transformation from the frequency domain to feature domain and effective semantic-style transformation for feature differences. We can summarize our work in the following way.

1) We propose a new thinking to explain the information composition of bitemporal HSIs. And based on this, we propose a frequency-domain information exchange and generation network with Siamese U-shaped structure (FDIEG-UNet) for optimal global joint spectral–spatial feature domain for HSI-CD.
2) We have designed an innovative FDGJLS module that can combine global feature learning and joint feature learning, and used it as a basis for information separation which provides basic information for subsequent processing of style feature removal.
3) We designed a ULDT module to transform the sum of obtained bitemporal invalid style information from the frequency domain to the feature domain to achieve the purpose of removal from the beginning. This provides the first pure semantic feature fusion method to optimize the recognition ability of unchanged samples using common information.
4) We have designed an SSTDFE module to extract temporal-related semantic difference features. This provides the second pure semantic feature fusion method to optimize the recognition ability of changed samples using difference information. Two modules work together to improve the equality of feature description between changed and unchanged samples in CD.

The remaining content is presented as follows. Section II provides an elaboration on related works regarding Fourier transformer in HSI-CD. Section III provides a detailed description of the FDIEG-UNet framework for HSI-CD. Section IV introduces the experimental results and comparative effects on three datasets. Finally, Section V is the conclusion.

II. Related Works

A. HSI-CD Based on Deep Learning

The deep learning-based method, with its powerful nonlinear representation capability, is currently the most commonly used change detection approach. In recent years, there has been a rise in DL techniques specifically designed for HSI-CD [16], [19], [23], [25], [28], [29], [30], [31], [32]. Due to the data structure of HSIs, high-resolution spatial information, and sequential spectral information, CNN [7] and Transformer [33] are increasingly favored by the HSI-CD industry. Wang et al. [28] proposed a method called GETNET. It performs spectral unmixing of linear and nonlinear features, fusing richer features by using CNN. Yang et al. [29] have proposed a deep multiscale pyramid network based on spatial–spectral residual attention enhancement. Ou et al. [34] have proposed a CNN framework to address spectral band redundancy by incorporating slow-fast band selection and feature fusion grouping. Chen et al. [19] combined temporal spatial spectral information through a transformer. Yu et al. [35] developed a band selection method to select a salient subset. Li et al. [36] proposed a fully convolutional network based on noise modeling for HSI-CD. Gong et al. [37] proposed a spatial attention mechanism based on distance using Gaussian sampling to address the impact of neighborhood spatial noise. Wang et al. [15] proposed a new transformer suitable for HSI-CD, which combines spatial information and spectral features. Based on integrating various deep learning structures, our FDIEG-UNet has been designed with a new foundational architecture.

Currently, some CD methods based on unsupervised and supervised learning have proposed a series of methods to address the issue of bitemporal images being affected by external factors due to different shooting times. Wiratama et al. [38] reused the previous feature maps through convolutional layers to measure the dissimilarity between two temporal images and improve the accuracy of CD. Lei et al. [39] proposed a differential enhancement and spatial spectrum non-local network to minimize the impact of irrelevant changes. Hou et al. [40] proposed a three-order tucker decomposition reconstruction detector for HSI-CD, which fully considers the intrinsic structural information of spectra. These methods use

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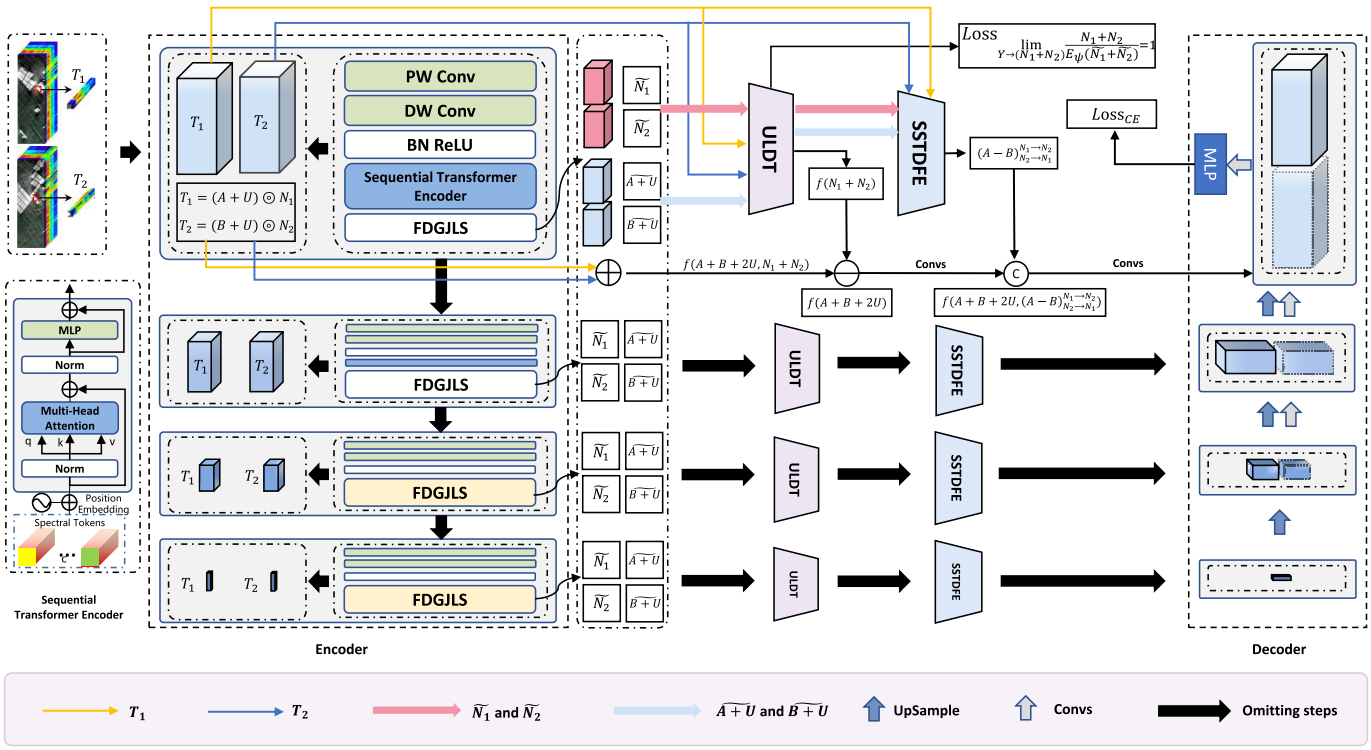


Fig. 2. Proposed framework of FDIEG-UNet. We first use an encoder to extract and combine spatial features and spectral sequence features. In the first two layers, the FDGJLS module only separates frequency-domain information, and in the last two layers, the FDGJLS module extracts joint global features and separates information. In residual connections, we use SSTDFE module and ULDT module to remove temporal-related common style features. Finally, the optimized joint features are multiscale extracted and predicted through a decoder.

domain adaptation or other methods to minimize the stylistic differences in images captured at different times.

B. Application of Fourier Transformer

In recent years, with the breakthrough of CNN in the field of computer vision, many studies have begun to integrate Fourier transform into DL methods to enhance the performance of visual tasks [27], [41], [42]. Some studies use discrete Fourier transform (DFT) to convert images to the frequency domain and utilize frequency-domain information to improve the performance of specific tasks [43]. Other studies replace the convolution in CNN with local Fourier unit, executing convolution in the frequency domain to accelerate CNN [44].

For HSIs, the information in the spectral domain is sequential and can be transformed into the frequency domain using Fourier transform to explore feature learning in the frequency domain. This helps enhance effective frequency-domain signals (corresponding to spectral features) and suppress noise [45], [46], [47].

However, this method for noise removal is rather rough. Currently, Rao et al. [48] have proposed a global filtering network to learn long-range spatial dependencies in the frequency domain after the Fourier transformation with logarithmic complexity, aiming to accomplish the task of learning interactions between spatial positions from raw data. Yang and Soatto [27] observed through the Fourier transform of images into the frequency domain that the amplitude describes the signal's contour and shape. The phase spectrum describes the position of the signal. Signals with the same shape at different positions have the same amplitude spectrum and different phase spectra. These two pieces of information can effectively describe both style and semantic features.

In this work, inspired by the current research on Fourier transform, we further separate and process information to remove the influence of invalid style information.

III. Proposed Method

Due to the fact that most current DL-based HSI-CD methods do not focus on the interdependence between style and semantic features in the spectral–spatial feature space. And simply using a crude bitemporal feature fusion strategy to represent the learned feature correlations ignores the equality and saliency of feature description between changed and unchanged samples. Inspired by this, we propose an FDIEG-UNet framework for HSI-CD. In this section, we provide the details of the proposed network.

A. Architecture

Our FDIEG-UNet architecture consists of three main components: 1) an encoder that performs temporally uncorrelated multiscale global joint spectral–spatial feature extraction and separates relevant dependency information; 2) residual connections, extracting temporally correlated semantic features with equality and saliency through our designed ULDT and SSTDFE modules; and 3) a decoder that learns and predicts from optimized temporally correlated multiscale feature subspaces. The entire process of our FDIEG-UNet is shown in Fig. 2.

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1) Encoder: Our model first uses an encoder with four layers of downsampling blocks for multiscale feature extraction. To adapt to the data structure of HSIs, we have redesigned an encoder that downsamples fine spatial and spectral features simultaneously. The downsampling of fine spatial features aims to compensate for the blurring and loss of edge and detail information in the detection regions caused by focusing only on semantic features in subsequent stages. The downsampling of spectral features aims to progressively eliminate redundant information in the spectra. As shown in Fig. 2, set T_1^0, $T_2^0 \in \mathbb{R}^{c\times p\times p}$ as the input bitemporal HSI patches, where p and c are the length and width of the patches, and the number of spectral channels in the HSIs, respectively. The encoder stage can be summarized as follows:

$$T_i^{j+1}, E_i^{j+1} = \begin{cases} \mathrm{FD}\left(\mathrm{TE}\left[f_{\mathrm{Convs}}\left(T_1^j; T_2^j\right)\right]\right), & j = 0, 1 \\ \mathrm{FD}\left[f_{\mathrm{Conv}}\left(T_1^j; T_2^j\right)\right], & j = 2, 3 \end{cases} \tag{3}$$

where TE$(\cdot)$ is sequential transformer encoder, and FD$(\cdot)$ is FDGJLS module. After first layer, output a $b \times c_1 \times (p-2) \times (p-2)$-sized features T_i^1. After downsampling at jth layer, output a $b \times (c_1/2^{j-1}) \times (p-2j) \times (p-2j)$-sized features T_i^j. E_1^{j+1} includes style information (amplitude) $\widetilde{N_1}$ and semantic information (phase components) $\widetilde{A+U}$ of the same size as T_i^{j+1} in frequency domain. E_2^{j+1} includes $\widetilde{N_2}$ and $\widetilde{B+U}$.

a) Learning of spatial features: To alleviate the problem of insufficient labeling in the HSI-CD dataset, we adopted a method that reduces parameters using depthwise separable convolution during downsampling at each layer. This approach leverages convolutional layers to obtain spatial feature mappings of input images. This unit contains pointwise convolutional layers and deep convolutional layers. After the convolutional layer, we added a normalization layer and a ReLU activation layer.

b) Learning of spectral sequence features: In the feature learning stage at the beginning of the encoder, due to the input of patches from the original HSI patches, which contain sufficient spectral sequence information, we have added a sequential transformer encoder (S-TE) in the first two layers of the encoder to generate enhanced global spectral sequence features. The architecture of S-TE is shown on the left side of Fig. 2, after passing through the downsampled convolutional layer, the bitemporal feature maps T_1', $T_2' \in \mathbb{R}^{c_j\times p_j\times p_j}$ are obtained. We reshape T_1' and T_2' to obtain token sets $T \in \mathbb{R}^{c_j\times p_j p_j}$. After position encoding, we use S-TE to model the context of these spectral sequences. Finally, we will reshape it to its original shape.

Generally speaking, S-TE is a structure that has three main modules: position embedding, self-attention (SA), and multilayer perceptron (MLP). The SA mechanism captures the internal correlations of spectral sequences by embedding the order of captured input sequences. The token sets T are input. The formula can be expressed as

$$\begin{aligned} T' &= \mathrm{MSA}(\mathrm{LN}(T)) + T \\ T &= \mathrm{MLP}(\mathrm{LN}(T')) + T' \end{aligned} \tag{4}$$

where LN is the normalization layer.

The input of the MSA is a triplet (query Q, key K, value V) calculated from the input token set T_i', where the MSA formula can be expressed as

$$\begin{aligned} Q &= TW^q \\ K &= TW^k \\ V &= TW^v \end{aligned} \tag{5}$$

$$\mathrm{Att}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{6}$$

$$\begin{aligned} \mathrm{MSA}(T) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^O \\ \mathrm{head}_j &= \mathrm{Att}\left(TW_j{}^q, TW_j{}^k, TW_j{}^v\right) \end{aligned} \tag{7}$$

where W^q, W^k, $W^v \in \mathbb{R}^{c\times d}$, $W^O \in \mathbb{R}^{hd\times c}$ are the learnable parameters of the three linear projection layers, h is the number of SA heads, and d represents the channel dimension of the triplet.

c) Learning and separation of global spectral–spatial features: HSI-CD is a task involving 3-D data, with a more complex data structure. Due to the input of the original data, there may be noise in both spatial and spectral dimensions, which can affect the learning of global features and greatly increase the computational complexity. Therefore, after obtaining high-level spectral and spatial features through the first two layers of feature learning, we then use the FDGJLS module for global joint feature learning. Our FDGJLS module separates relevant dependency information in the frequency domain. As shown in Fig. 2, the FDGJLS module is represented by white for no learnable parameters and beige for feature learning. After completing spectral sequence feature learning or downsampling, the feature maps T_1', $T_2' \in \mathbb{R}^{c_j\times p_j\times p_j}$ are input into the FDGJLS module. We first convert to the frequency domain, then learn, separate, and finally invert transform to obtain features T_1, T_2, E_1, and E_2 of the same size as the input.

Our key is to learn before separating, whether it is the first two layers or the last two layers. We enable feature learning to adaptively reduce the invalid style information that has less impact on CD, allowing feature extraction to obtain the most saliency high-level semantic information. At the same time, to ensure that the separated frequency-domain information better fits the features to be processed in residual connections.

2) Residual Connection: After obtaining the output features of the encoder, we input all features separately into the SSTDFE module and ULDT module for temporal-related feature subspace processing and learning. Based on the features obtained from the two modules, the original features $T_1 + T_2$ and $T_1 - T_2$ are combined in different ways to learn semantic features with saliency and equality. Finally, we obtain optimized feature maps of the same size as T_i. It is beneficial for the comparative learning feedback of dual temporary features.

We need the representation of N_1+N_2 in the feature domain, which allows for better subtraction to achieve the removal of invalid style features. On the other hand, $A - B$ does not require representation in the feature domain because the combination of features is simpler. And we use concatenation instead of addition to combine the two features to prevent $A - B$ conflicts with $A + B$.

The reason for first removing and then combining here is to prevent the two temporal-related semantic features from

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affecting the structure or information of the bitemporal style features through concatenation and convolutional layers during the learning and combination process. This approach ensures that the removal of invalid style information is effective without the need to relearn the data structure, which could otherwise slow down the convergence speed.

We detail the inference procedure of our residual connection in Algorithm 1.

Algorithm 1 Residual Connection

Input:

$\mathcal{T} = \{T_1^j, T_2^j\}$ (Features of the j-th encoder layer)

$\mathcal{X} = \{X^{j-1}\}$

Output:

$\mathcal{X} = \{X^j\}$ (Features of the j-th decoder layer)

// step1: extracting joint features and separate information

while $i = 1, 2$ **do**

$T_i^j, \widetilde{N_i}, \widetilde{A+U}, \widetilde{B+U} = \text{Encoder}_j(T_i^{j-1})$

end while

// step2: temporal related feature subspace processing

$N_1 + N_2 = \text{ULDT}_j(T_i^j, \widetilde{N_i}, \widetilde{A+U}, \widetilde{B+U})$

$A - B = \text{SSTDFE}_j(T_i^j, \widetilde{N_i}, \widetilde{A+U}, \widetilde{B+U})$

// step3: extracting semantic features with equality and saliency

$T_s^j = \text{Convs}(T_1 + T_2 - (N_1 + N_2))$

$T_{se}^j = \text{Convs}(\text{Concat}(T_s, A - B))$

// step4: optimized multi-scale feature extraction

$X = \text{Concat}(T_{se}^j, X^{j-1})$

$X^j = \text{Decoder}_j(X)$

3) Decoder: Our model uses a decoder with three layers of upsampling and convolutional layers to learn and predict the optimized multiscale temporal-related feature subspace. By continuously upsampling to restore resolution, predictions become more accurate in detecting subtle changes at multiple scales. The encoder stage can be summarized as follows:

$$X^j = \begin{cases} f_{\text{Convs}}\left[\text{Concat}\left(T_{\text{se}}^j, \text{UP}\left(X^{j+1}\right)\right)\right], & j = 1, 2 \\ f_{\text{Convs}}\left[\text{Concat}\left(T_{\text{se}}^j, \text{UP}\left(T_{\text{se}}^{j+1}\right)\right)\right], & j = 3 \end{cases} \tag{8}$$

where j represents the top-down jth layer in both the encoder and decoder, T_{se}^j is the features obtained from the jth layer residual connection, and X^j is the features obtained from the jth layer decoder. After jth layer of the decoder, output a $b \times (c_1/2^{j-2}) \times (p - 2j) \times (p - 2j)$-sized features X^j.

The X^1 is fed into the prediction header for pixel-level prediction. The prediction head formula is as follows:

$$M = \sigma(g(X^1)) \tag{9}$$

where $g(\cdot)$ represents a fully connected layer, and $\sigma(\cdot)$ represents the softmax activation function.

B. Frequency-Domain Global–Joint Learning and Separation Module

Our process involves transforming information from the feature domain to the frequency domain, learning in the frequency domain, and finally separating information based on the characteristics of the frequency domain. In order to better separate information, we introduce FT to obtain two types of frequency-domain information, providing basic information for subsequent style and semantic feature extraction. However, in HSI, different objects may exhibit similar spectral features, while the same object may exhibit different spectral features at different positions. The semantic and style information we separate in the frequency domain also needs to consider the spatial–spectral information comprehensively. So before separating features, we need to transform the entire feature space rather than separately transforming individual spatial or spectral dimensions, otherwise it will be difficult to jointly consider spatial–spectral information during separation.

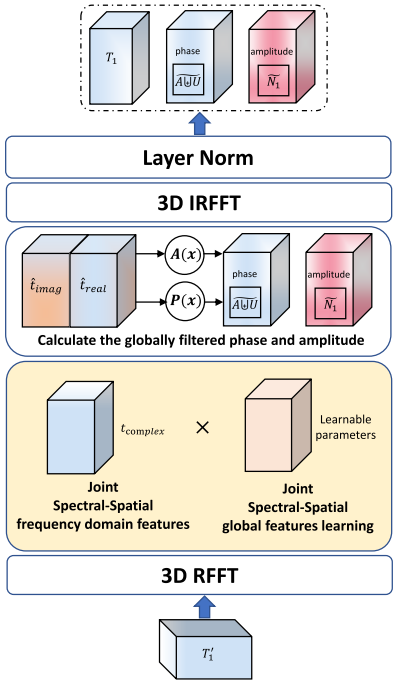


Fig. 3. FDGJLS module.

Inspired by GFNet's research on self-attention of 2-D spatial FT [48], we have redesigned a module that can learn the interaction between spatial and spectral positions in the frequency domain. Introducing the 3-D DFT, we use learnable parameters to act on the spectrum converted by the 3-D features for learning and modeling. Unlike VIT [33], which can only capture long-term dependencies in one spatial or spectral dimension, and 3-D CNN, which can only combine local features, our advantage lies in using 3-D FT to mix information between spectral and spatial tokens, satisfying not only global feature learning akin to the VIT feature domain but also spatial–spectral joint feature learning akin to 3-D CNN, thereby better executing global joint information separation.

The DFT plays a crucial role in our network, and here we provide the formula for converting a sequence into a frequency domain using 1-D DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} := \sum_{n=0}^{N-1} x[n] W_N^{kn} \tag{10}$$

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where j is the imaginary unit, $x[n]$ is a sequence of N complex numbers, $0 \le n \le N-1$. $W_N = e^{-j(2\pi/N)}$, and $X[k]$ is the spectrum of the sequence $x[n]$ at frequency $\omega = 2\pi k/N$.

The formula for inverse DFT (IDFT) is

$$x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]e^{j(2\pi/N)kn}. \tag{11}$$

Given 3-D signals $x[m,n,k], 0 \le m \le M-1, 0 \le n \le N-1, 0 \le k \le K-1$. The above DFT can be extended to 3-D DFT

$$X[u,v,w] = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\sum_{k=0}^{K-1} x[m,n,k]e^{-j2\pi\left(\frac{um}{M}+\frac{vn}{N}+\frac{wk}{K}\right)}. \tag{12}$$

Through the property of conjugate symmetry, we can see that half of the spectrum $X[u,v,w]$ contains all the information needed for $x[m,n,k]$. Therefore, we remove the symmetric part here to eliminate redundancy and improve computational efficiency.

As shown in Fig. 3, set T_1' as the input. Set T_1, $\widetilde{N_1}$, $\widetilde{A+U}$ as the output. Obtaining frequency-domain information $t_1 \in \mathbb{C}^{c_j \times p_j \times \lceil p_j/2 \rceil}$ through unprocessed 3-D features through 3-D fast FT (FFT). Here, we set a learnable parameter $K \in \mathbb{R}^{c_j \times p_j \times \lceil p_j/2 \rceil \times 2}$. After converting it to a complex number, we modulate the spectrum by element-wise multiplication. Perform global adaptive filtering of style features in the same dimension for global joint feature learning. The above formula is expressed as

$$\hat{t}_i = K \odot t_i = K \odot \mathcal{F}\left[T_i'\right] \tag{13}$$

where $\odot$ is the element-wise multiplication and $\mathcal{F}[\cdot]$ denotes the 3-D FFT.

And, we use the following formula to extract relevant information (amplitude and phase components):

$$A[u,v,w] = \left[\hat{t}_{\text{real}}[u,v,w]^2 + \hat{t}_{\text{imag}}[u,v,w]^2\right]^{\frac{1}{2}} \tag{14}$$

$$P[u,v,w] = \arctan\left[\frac{\hat{t}_{\text{imag}}[u,v,w]}{\hat{t}_{\text{real}}[u,v,w]}\right] \tag{15}$$

where $A[u,v,w]$ is amplitude $\widetilde{N_1}$, $\widetilde{N_2}$ and $P[u,v,w]$ is phase components $\widetilde{A \uplus U}$, $\widetilde{B \uplus U}$.

Then, the spectrum is transformed into the spatial–spectral feature domain through 3-D IFFT

$$T_i = \mathcal{F}^{-1}[\hat{t}_i] \tag{16}$$

where $\mathcal{F}^{-1}[\cdot]$ denotes the 3-D IFFT.

Finally, we pass all the obtained information through a normalization layer to prevent gradient explosion caused by the multiplication of parameters in the frequency domain.

C. Unsupervised Learnable Domain Transformation Module

It can be known that the representation of style information in the frequency domain and feature domain is different. But they contain the same information, which can be represented as follows:

$$\widetilde{N_1} + \widetilde{N_2} = \mathcal{F}_A(f(T_1)) + \mathcal{F}_A(f(T_2)) \tag{17}$$

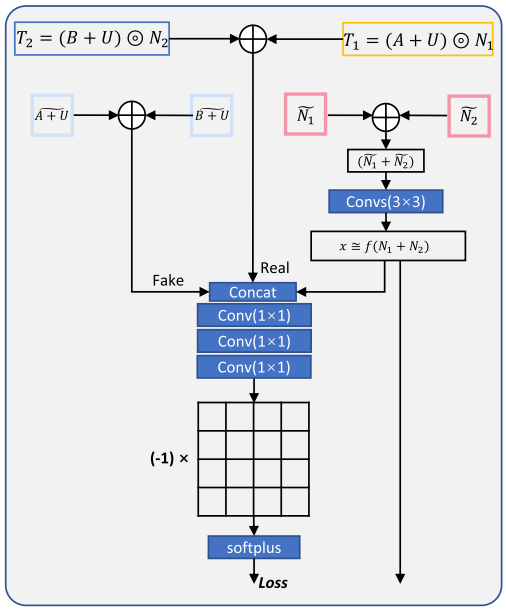


Fig. 4. ULDT module.

where $f(\cdot)$ is the representation of style or semantic information in the feature domain, and $\mathcal{F}_A(\cdot)$ is the process of extracting amplitude in the frequency domain through FFT.

We need to use the sum of invalid style information in the feature domain to optimize the feature domain in residual connections. However, we only know the representation of the two invalid style features in the frequency domain, and direct subtraction cannot remove them. Instead, it can cause confusion and redundancy of invalid style information, as well as the problem of data information disorder caused by the combination of frequency domain and feature domain.

As shown in Fig. 4, we designed the ULDT module. This module studies unsupervised learning of style representation transformation by maximizing the mutual information between the features $I_a^+ = T_1 + T_2 = f(A+B+2U, N_1+N_2)$ and $E_\psi(I_t^- = \widetilde{N_1} + \widetilde{N_2})$ in the feature domain. Setting $I_e^- = \widetilde{A+U} + \widetilde{B+U}$. Inspired by contrastive learning, we flexibly formulate representation learning objectives for specific targets through a discriminator and encoder by forcibly constraining the same invalid style information contained in two distributions, achieving the transformation of learning from frequency domain to feature domain.

Here, we choose I_a^+ as the positive sample of the feature domain, and I_e^- as the negative sample of the frequency-domain semantic sum. We use concatenation in the channel dimension to form positive sample pairs of $(E_\psi(I_t^-), I_a^+)$, and negative sample pairs of $(E_\psi(I_t^-), I_e^-)$. After concatenation, we use three consecutive $\text{Convs}^{1\times1}$ to change the number of channels from $2 \times c_j$ to c_j, and then to 1. These three convolutional layers are parameterized discriminators that estimate the degree of mutual information. The formula we use for maximizing Jensen-Shannon MI estimation [49] is as follows:

$$\hat{\mathcal{I}}^{\text{JSD}}_{\omega,\psi}\left(E_\psi\left(I_t^-\right), I_a^+\right) := \mathbb{E}_{\mathbb{P}}^{f\times\mathbb{F}}\left[-\text{sp}\left(-T_{\omega,\psi}\left(E_\psi\left(I_t^-\right); I_a^+\right)\right)\right] - \mathbb{E}_{\mathbb{P}\times\tilde{\mathbb{P}}}^{\mathbb{F}}\left[\text{sp}\left(T_{\omega,\psi}\left(\psi\left(I_t^-\right); I_e^-\right)\right)\right] \tag{18}$$

$$\text{sp}(x) = \log\left(1+e^x\right) \tag{19}$$

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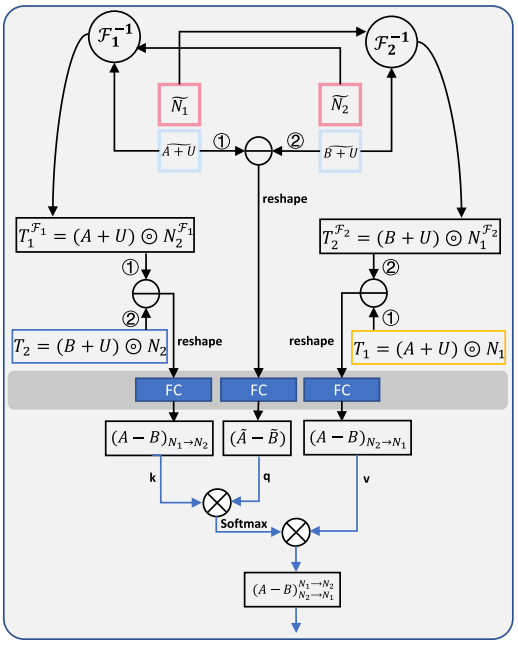


Fig. 5. SSTDFE module.

where T_ω is a discriminator function modeled by convs with parameters ω, E_ψ is a convolutional layer encoder with parameters ψ used for learning to transform frequency domain to feature domain, $\mathbb{P}$ is the distribution of invalid style information, $\widetilde{\mathbb{P}}$ is the distribution of semantic information, f is the distribution of feature domains, $\mathbb{F}$ is a frequency-domain distribution, and $\mathrm{sp}(\cdot)$ is the softplus activation function.

Finally, we can obtain the loss function

$$L_{\mathbb{F}\to f}(\hat{\omega},\hat{\psi}) = \mathrm{argmax}_{\omega,\psi}\frac{1}{M^2}\sum_{i=1}^{M^2}\hat{\mathcal{I}}_{\omega,\psi}\left(E_\psi\left(I_t^-\right), I_a^+\right) \quad (20)$$

where the feature size is $M = p_j \times p_j$.

To minimize the loss function, we multiply $L_{\mathbb{F}\to f}(\hat{\omega},\hat{\psi})$ by -1. Maximizing mutual information during the convergence process to transform potential structures and representations in the data. Make $E_\psi(I_t^-)$ far away from the frequency domain and the sum of semantic information, close to the feature domain and the sum of invalid style information. Finally, obtain the sum of invalid style features close to the feature domain, expressed as follows:

$$E_\psi(I_t^-) \cong f(N_1 + N_2). \quad (21)$$

D. Semantic-Style Transformation Difference Feature Extractor

We need to extract semantic differentiation features for optimizing the feature domain in residual connections. Just a simple $T_1 - T_2 = f(A-B, N_1 - N_2)$ will have the influence of invalid style information. As shown in Fig. 5, the SSTDFE module uses a simple method to adapt images from different time series to different invalid styles. By exchanging amplitude information from different time series in the frequency domain, the difference between the source distribution and the target distribution is reduced. Without changing the semantic content, the source features are mapped to the target style. And obtain bitemporal feature information that adapts to different styles in the feature domain through IFFT. The above can be expressed as

$$T_{s\to t}^{\mathcal{F}} = \mathcal{F}^{-1}[\mathcal{F}_A(T_t), \mathcal{F}_P(T_s)] \quad (22)$$

where s and t represent two time series, respectively, and $s \neq t$. $\mathcal{F}_A(\cdot)$ is the amplitude and $\mathcal{F}_P(\cdot)$ is the phase, both extracted in the frequency domain through FFT. We can obtain $T_1^{\mathcal{F}_1} = f(A+U, N_2^{\mathcal{F}_1})$, $T_2^{\mathcal{F}_2} = f(B+U, N_1^{\mathcal{F}_2})$.

The feature-domain information is obtained through IFFT after amplitude exchange, in which the style information will undergo a certain transformation based on semantic features, $N_s^{\mathcal{F}_t} \neq N_s$. These transformations are different for changing and unchanged pixels, so they are beneficial for CD and we reserve them. Then, we use simple subtraction to remove the influence of source invalid style information. As shown in Fig. 5, note that the subtraction here is sequential

$$\begin{aligned} T_1^{\mathcal{F}_1} - T_2 &= f\left(A+U, N_2^{\mathcal{F}_1}\right) - f(B+U, N_2) \\ T_1 - T_2^{\mathcal{F}_2} &= f(A+U, N_1) - f\left(A+U, N_1^{\mathcal{F}_2}\right) \\ \widetilde{A+U} - \widetilde{B+U} &= \widetilde{A} - \widetilde{B}. \end{aligned} \quad (23)$$

Finally, we use a fully connected layer to learn $\widetilde{A} - \widetilde{B}$, $(A-B)_{N_1\to N_2}$, $(A-B)_{N_2\to N_1}$. We combine this saliency information in the frequency domain and feature domain using an attention mechanism to obtain saliency semantic differentiation features. The formula is as follows:

$$\begin{aligned} Q &= (\widetilde{A} - \widetilde{B})W^q \\ K &= (A-B)_{N_1\to N_2}W^k \\ V &= (A-B)_{N_2\to N_1}W^v \end{aligned} \quad (24)$$

$$(A-B)_{N_2\to N_1}^{N_1\to N_2} = \sigma(Q\otimes K)\otimes V \quad (25)$$

where $\sigma(\cdot)$ represents the softmax activation function, and $\otimes$ is matrix multiplication.

E. Loss Function

Finally, during the training phase, we use cross-entropy loss and $L_{\mathbb{F}\to f}(\hat{\omega},\hat{\psi})$ to optimize network parameters. The calculation formula for the cross-entropy loss is as follows:

$$L_{\mathrm{CE}} = -\frac{1}{n}\sum_{i=1}^{n}(y_i \log\hat{y}_i + (1-y_i)\log(1-\hat{y}_i)) \quad (26)$$

where n represents the number of samples, y_i represents the true label, and $\hat{y}_i$ represents the predicted probability value calculated by the softmax function.

The calculation formula for the loss function is as follows:

$$L = L_{\mathrm{CE}} + \lambda L_{\mathbb{F}\to f}(\hat{\omega},\hat{\psi}). \quad (27)$$

IV. Experiments

In this section, we analyze our proposed FDIEG-UNet through experiments. First, three datasets were described. Second, the evaluation criteria, experimental environment, and experimental setup were introduced. Third, compare the proposed method with other state-of-the-art HSI-CD algorithms. Fourth, through the analysis of ablation experiments, the innovative points of the proposed method are revealed. Fifth, analyze the hyperparameters of FDIEG-UNet.

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Fig. 6. Three HSI-CD datasets (Farmland dataset, Santa Barbara dataset, and Bay Area dataset).

A. Datasets

1) Farmland Dataset: The bitemporal HSIs in the dataset were obtained in 2006 and 2007, respectively. The size of each image is 420 × 140 pixels, which covers some of the changes in arable land in Yancheng City, Jiangsu Province, China. After preprocessing, the HSIs contain a total of 155 spectral bands. Farmland dataset includes 40 417 unchanged pixel samples and 18 383 changed pixel samples.

2) Santa Barbara Dataset: The bitemporal HSIs from the Santa Barbara dataset we used were collected using AVIRIS sensors in 2013 and 2014, respectively, to capture partial variation areas in Santa Barbara, California. Each HSI contains 984 × 740 pixels, including 80 418 unchanged pixel samples, 52 134 changed pixel samples, and 595 608 pixel samples with unknown states. After preprocessing, the HSIs contain a total of 255 spectral bands.

3) Bay Area Dataset: The bitemporal HSIs in this dataset were also collected using AVIRIS sensors in 2013 and 2014, respectively, to capture the changed pixels in the Patterson area of California. Each HSI contains 600 × 500 pixels, including 34 211 unchanged pixel samples, 38 425 changed pixel samples, and 227 364 pixels with unknown states. After preprocessing, the HSIs contain a total of 255 spectral bands.

As shown in Fig. 6, we visualize three datasets. We divide the samples into training, validation, and testing sets, and partition the number of samples for each subset. We construct the training and validation sets by selecting changing and unchanged samples in a 1:1 ratio. To compare with other models on a limited sample size, we select 2% of the samples for the training sets. Because we only selected 2% of the data for training, the selection of data will have an impact on the experiment. In the following experiments, we will randomly select samples for five experiments. As shown in Table I, we take the average value of the experiment, and the subscript of each data is the standard error.

B. Experimental Setup

1) Implementation Details: FDIEG-UNet was implemented utilizing an Intel® Xeon[1] Silver 4210R CPU running at 2.40 GHz with 40 cores and the NVIDIA GeForce RTX 3090 GPU within PyTorch and MindSpore. Stochastic gradient descent (SGD) is employed with a momentum of 0.90 and a weight decay of 0.0005 for model optimization. The initial learning rate is 0.0005, and then every 1/20 of the total epochs is multiplied by 0.9 during the 200 epochs of training. The patch size is 9, the batch size is 16, and there is one sequential sequence encoder integrating positional encoding.

2) Evaluation Metrics: In this study, we used overall accuracy (OA) and kappa coefficient (KC) as indicators to evaluate model performance. KC measures the agreement between two raters. The calculation formula is as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN}} \tag{28}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{29}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{30}$$

$$F1 = \frac{2\times \mathrm{Pr}\times \mathrm{Re}}{\mathrm{Pr}+\mathrm{Re}} \tag{31}$$

$$Pc = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})}{(\mathrm{TP}+\mathrm{FP}+\mathrm{TN}+\mathrm{FN})^2} \tag{32}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-Pc}{1-Pc} \tag{33}$$

where true positive (TP) represents the changed pixels that are correctly identified in the classifier's predictions, true negative (TN) represents the unchanged pixels that are correctly identified in the classifier's predictions, false positive (FP) denotes the unchanged pixels incorrectly identified as changed in the classifier's predictions, and false negative (FN) indicates the changed pixels that are incorrectly identified as unchanged in the classifier's predictions.

C. Comparison to State-of-the-Art

To evaluate the performance of the proposed method, we further compared our method with eight other state-of-the-art algorithms. Some of the classic HSI-CD algorithms include CVA [50], PCA [51], and SVM. To facilitate the comparison with some classic DL algorithms, we transferred some remote sensing image algorithms and implemented some algorithms adapted to HSI-CD, including GETNET [28], SAE + FCM + CNN (SF-CNN) [7], BIT-CD [52], SST-Former [15], SATNet [25], DCENet [53], and STSS-Net [19]. The details of these algorithms are as follows.

CVA is a commonly used algebraic method that calculates the Euclidean distance of all pixels to obtain change vectors, and finally classifies them using K-means. We set the patch size to 5 in the comparative experiment.

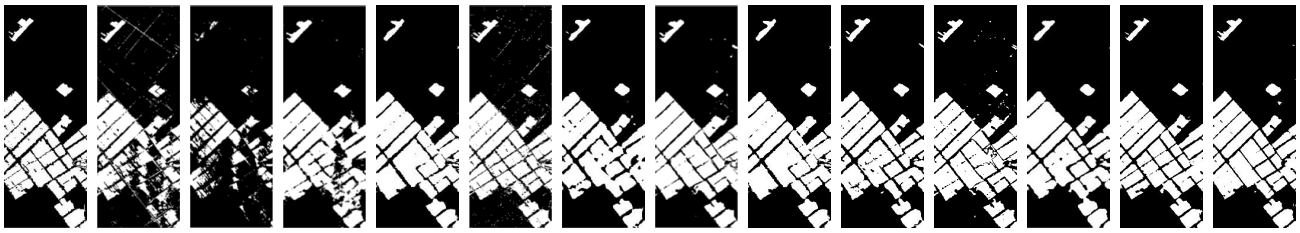
PCA named differential principal component analysis, which maps the raw data to a low-dimensional feature space

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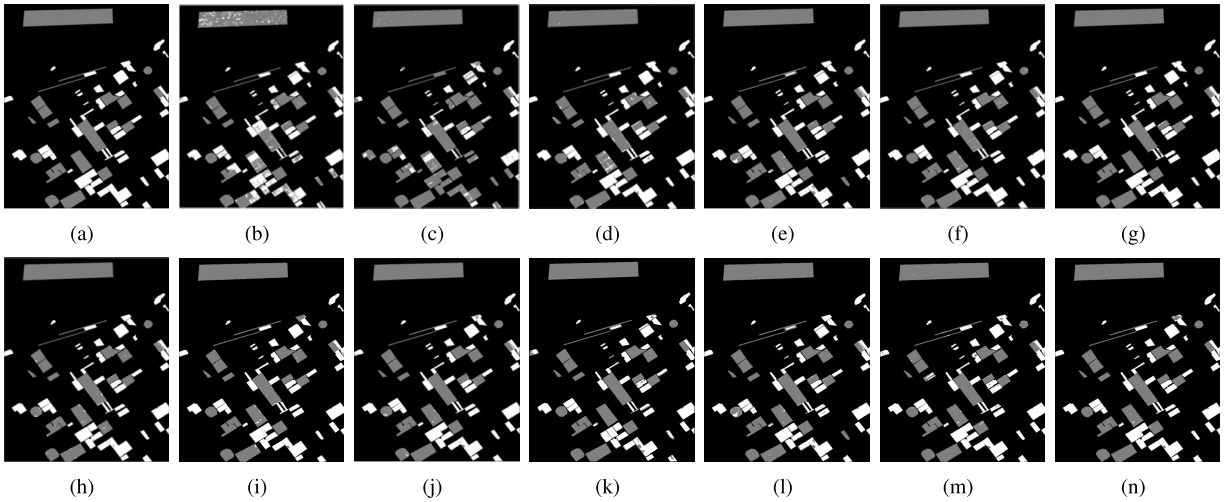
TABLE I
COMPARISON OF TEST RESULTS OF EXCELLENT HSI-CD METHOD ON THREE DATASETS

Method	Farmland					Santa Barbara					Bay Area				
	OA%	KC	F1%	Pr%	Re%	OA%	KC	F1%	Pr%	Re%	OA%	KC	F1%	Pr%	Re%
CVA	$87.49_{1.13}$	$69.98_{3.53}$	$80.26_{2.18}$	85.88	75.33	$83.25_{1.93}$	$65.19_{3.11}$	$79.15_{2.56}$	77.47	80.91	$85.17_{1.35}$	$70.63_{2.21}$	$84.40_{1.98}$	90.16	79.33
PCA	$81.00_{5.13}$	$48.09_{8.82}$	$66.98_{7.42}$	92.96	52.35	$75.66_{6.12}$	$44.03_{9.25}$	$78.17_{8.31}$	72.73	84.50	$67.77_{9.18}$	$37.90_{12.1}$	$58.57_{10.3}$	**99.21**	41.55
SVM	$90.87_{1.65}$	$78.97_{2.13}$	$84.64_{1.98}$	76.84	94.20	$97.01_{1.02}$	$93.77_{2.35}$	$96.78_{1.08}$	97.37	96.20	$90.51_{1.51}$	$80.94_{2.68}$	$86.82_{1.60}$	90.95	83.05
SF-CNN	94.48	88.63	/	/	/	95.35	90.12	/	/	/	95.81	91.59	/	/	/
GETNET	$94.03_{1.03}$	$86.40_{2.98}$	$90.60_{1.41}$	92.04	89.21	$97.37_{0.75}$	$94.48_{2.29}$	$97.02_{1.18}$	96.44	97.60	$97.49_{0.73}$	$94.72_{1.89}$	$96.86_{0.91}$	96.53	97.20
BIT-CD	$95.14_{0.72}$	$88.75_{2.03}$	$91.83_{0.98}$	90.63	93.07	$98.50_{0.77}$	$96.85_{1.89}$	$98.17_{0.88}$	98.61	97.73	$97.10_{0.42}$	$94.19_{1.51}$	$97.02_{0.55}$	96.44	97.60
SST-Former	95.21	88.43	/	/	/	98.10	96.58	/	/	/	97.89	95.94	/	/	/
SATNet	$95.32_{1.04}$	$80.10_{1.97}$	$85.77_{1.75}$	**94.24**	78.69	$98.19_{1.10}$	$88.23_{1.97}$	$91.83_{1.25}$	90.63	93.07	$96.12_{1.73}$	$84.03_{2.20}$	$89.02_{1.89}$	88.34	89.72
STSS-Net	$96.20_{0.29}$	$91.16_{0.47}$	$93.92_{0.39}$	93.94	93.90	$98.56_{0.23}$	$96.97_{0.30}$	**98.82**$_{0.28}$	98.53	**99.10**	$98.20_{0.25}$	$96.40_{0.36}$	$98.17_{0.35}$	97.92	98.42
DCENet	$95.00_{0.61}$	$88.53_{1.73}$	$92.21_{0.88}$	89.87	94.67	$98.03_{0.53}$	$86.27_{1.21}$	$87.34_{0.71}$	90.45	84.43	$94.17_{1.32}$	$76.01_{4.01}$	$79.40_{2.61}$	83.0	76.11
U-Net	$95.31_{0.43}$	$89.14_{0.86}$	$92.56_{0.45}$	90.89	94.30	$97.58_{0.31}$	$94.90_{0.45}$	$96.36_{0.39}$	97.19	95.55	$97.00_{0.33}$	$93.65_{0.52}$	$95.11_{0.44}$	96.25	94.00
Base+	$96.03_{0.35}$	$90.82_{0.51}$	$93.65_{0.46}$	92.52	94.80	$98.03_{0.24}$	$95.88_{0.44}$	$97.50_{0.32}$	97.46	97.54	$97.62_{0.38}$	$95.31_{0.71}$	$96.58_{0.49}$	96.39	96.78
Proposed	**96.50**$_{0.19}$	**92.28**$_{0.30}$	**94.57**$_{0.21}$	92.83	**96.37**	**98.82**$_{0.18}$	**97.50**$_{0.22}$	$98.46_{0.32}$	**99.64**	97.32	**98.55**$_{0.16}$	**97.10**$_{0.20}$	**98.65**$_{0.21}$	98.70	**98.60**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n)

Fig. 7. CD results on the Farmland dataset. (a) Ground truth. (b) CVA. (c) PCA. (d) SVM. (e) SF-CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) SATNet. (j) STSS-Net. (k) DCENet. (l) U-Net. (m) Base. (n) FDIEG-UNet. The white area is changed, and the black area is unchanged.



(a) (b) (c) (d) (e) (f) (g)

(h) (i) (j) (k) (l) (m) (n)

Fig. 8. CD results on the Santa Barbara dataset. (a) Ground truth. (b) CVA. (c) PCA. (d) SVM. (e) SF-CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) SATNet. (j) STSS-Net. (k) DCENet. (l) U-Net. (m) Base. (n) FDIEG-UNet. The white is the changed area. The gray is the unchanged area. Black is the unknown area.

and selects key features for classification through k-means. We set the patch size to 5 in the comparative experiment.

SVM, also known as support vector machine, is a supervised method that generates high-dimensional features for classification using the RBF algorithm.

SF-CNN, a CD method that combines CNN, sparse autoencoder, and unsupervised clustering.

GETNET performs hyperspectral unmixing, constructs affinity matrix, and further extracts features through CNN. We use the same other parameters as the original article to train GETNET.

BIT-CD improves the accuracy of CD by extracting high-level spatial semantic features. However, the backbone is not suitable for HSI, so we modified the Resnet [54] to the same depth separable convolution as in this article, while retaining the architecture of the original model. We adapt it to the input size of HSI data and set the batch size to 5.

SST-Former has designed a temporal-related transformer architecture for HSI-CD.

SATNet explores temporal-related feature fusion methods from different perspectives. The parameters are consistent with the original article.

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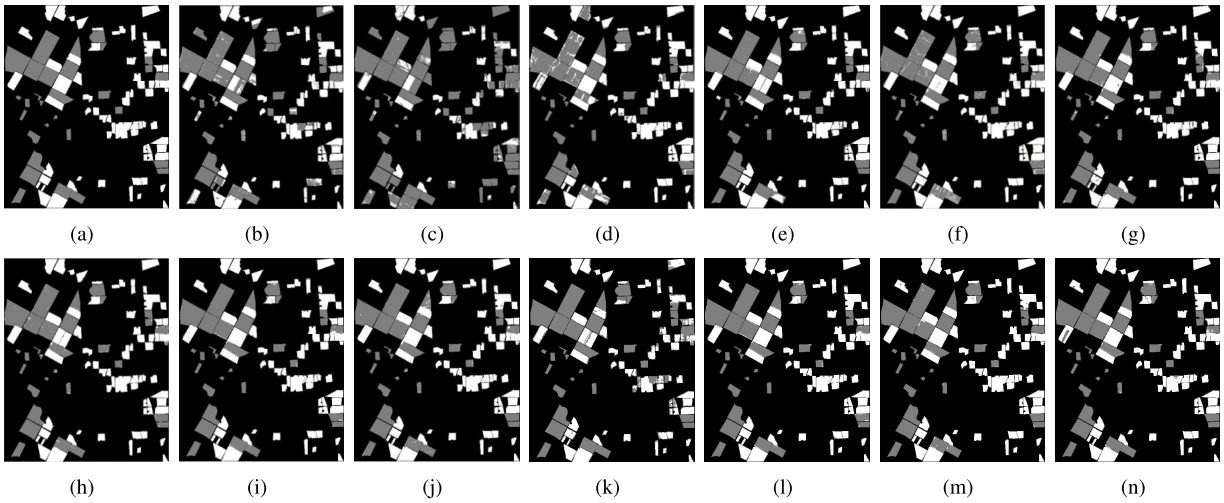


Fig. 9. CD results on the Bay Area dataset. (a) Ground truth. (b) CVA. (c) PCA. (d) SVM. (e) SF-CNN. (f) GETNET. (g) BIT-CD. (h) SST-Former. (i) SATNet. (j) STSS-Net. (k) DCENet. (l) U-Net. (m) Base. (n) FDIEG-UNet. The white is the changed area. The gray is the unchanged area. Black is the unknown area.

TABLE II
COMPARISON OF TEST RESULTS WITH DIFFERENT SAMPLE SIZES AND COMPARISON OF COMPUTATIONAL COMPLEXITY OF DIFFERENT MODELS

Dataset	Samples	BIT-CD			SATNet			STSS-Net			DCENet			Proposed		
		OA%	KC	F1%	OA%	KC	F1%	OA%	KC	F1%	OA%	KC	F1%	OA%	KC	F1%
Farmland	2%	95.14	88.75	91.83	95.32	80.10	85.77	96.20	91.16	93.92	95.00	88.53	92.21	**96.50**	**92.28**	**94.57**
Santa Barbara		98.50	96.85	98.17	98.19	88.23	91.83	98.56	96.97	**98.82**	98.03	86.27	87.34	**98.82**	**97.50**	98.46
Bay Area		97.10	94.19	97.02	96.12	84.03	89.02	98.20	96.40	98.17	94.17	76.01	79.40	**98.55**	**97.10**	**98.65**
Farmland	5%	96.10	90.75	92.84	93.35	84.76	89.66	**96.83**	**92.41**	**95.14**	96.21	91.24	94.01	96.65	92.53	94.64
Santa Barbara		98.52	96.98	98.32	98.21	88.55	89.20	98.78	97.40	**98.72**	98.17	87.31	88.30	**98.92**	**97.70**	98.58
Bay Area		97.24	94.37	97.19	96.65	84.48	86.62	98.31	97.05	98.27	96.31	92.41	96.79	**98.86**	**97.56**	**98.64**
Farmland	10%	96.83	92.11	94.88	95.30	79.33	81.77	**97.27**	**93.64**	**95.62**	96.99	93.02	95.21	96.61	92.45	94.66
Santa Barbara		98.50	96.88	98.30	98.39	88.83	89.69	98.81	97.62	98.75	98.40	88.13	89.21	**98.90**	**97.68**	**98.78**
Bay Area		97.78	95.74	97.67	96.30	84.48	86.62	98.36	97.01	98.45	94.17	77.33	80.76	**98.78**	**97.40**	**98.57**
Complexity (FLOPs)		4.35G			786.727M			289.434M			16.658G			939.839M		
Parameters		3.55M			1.364M			534.948K			16.006M			1.886M		

STSS-Net extracts salient high-level semantic features by attention mechanism and spectral decorrelation, and then performs temporal-related feature fusion.

DCENet is a diff-feature contrast enhancement network for HSI CD. To ensure fairness in data volume, we use the same amount of labeled data to train DCENet.

U-Net [55] is a pure transformer network with a Siamese U-shaped structure.

Base+, our baseline is Siamese U-Net [55] and we optimize the architecture based on it. It consists of an encoder (FDGJLS and S-TE), an unoptimized residual connection, and decoder.

We compared the proposed method with the above methods. However, due to some papers not disclosing the code and parameters, these indicators were not analyzed in the article. We only retain some comparable methods. Table I displays the comparison results of the evaluation indicators for CD across three datasets. Figs. 7–9 show the visual comparison results of CD for three datasets, respectively.

1) Quantitative Results: We can see that we achieved better results with limited samples. Compared to STSS-Net, we achieved better performance with similar standard errors. According to the standard error, it has been proven that the proposed method is more stable than other methods based on limited samples. Quantitative results show that on three datasets, our FDIEG-UNet achieved higher OA, Kappa, and F1-scores compared to the state-of-the-art methods we compared against. Furthermore, compared to the baseline, our FDIEG-UNet KC scores improved by 1.46/1.62/1.79. The OA scores exceeded the baseline scores by 0.47/0.79/0.93. The $F1$-scores exceeded the baseline scores by 0.92/0.96/2.07. This reflects the effectiveness of our FDIEG-UNet and the proposed modules in the HSI-CD task.

In comparison to unsupervised algorithms, some classic HSI-CD machine learning methods such as CVA, PCA, and SVM still exhibit significant performance gaps compared to DL-based methods in the HSI-CD task. While supervised SVM has shown significant improvements, it still performs poorly in nonlinear feature extraction compared to DL-based methods. It also indicates that the CVA and PCA algorithms misjudge a large number of invariant regions into changing

TABLE III
ANALYSIS OF THE COMPUTATIONAL COMPLEXITY OF THE PROPOSED ALGORITHM

	Complexity (FLOPs)	Parameters
Depthwise Convolution	$O(k^2HWD)$	k^2D
Attention	$O(HWD^2 + H^2W^2D)$	$4D^2$
FDGJLS	$O(HWD[log_2(HWD)] + HWD)$	HWD

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regions on the Santa Barbara dataset, thus having a high Re but a low KC.

The methods for optimizing HSI-CD can be divided into optimization of feature extraction and feature fusion.

In the optimization of feature extraction, compared to the latest STSS-Net, KC was improved by 1.12/0.63/0.7. The OA scores exceeded the latest STSS-Net by 0.3/0.26/0.35. The F1-scores exceeded the latest STSS-Net by 0.65/−0.36/0.48. We can see that we have greatly improved our KC score, combined with the improvement in Pr and Re scores, which is particularly effective in distinguishing between changing and non-changing samples. At the same time, in subsequent ablation experiments, we will demonstrate the improvement of recognition efficiency for changed samples. In comparison to some methods like SF-CNN, GETNET, BIT-CD, and STSS-Net, which optimize feature extraction based on noise modeling, we propose a new approach based on interpreting bitemporal HSI information composition, using a learnable method to address known interference variability from the outset. Compared with the semi-supervised method of DCENet, we achieved better results with the same amount of labeled data. We are missing 10% of unlabeled data compared to DCENet, but we have achieved better results. By optimizing the composition of information in the feature domain to enhance CD performance, this does not include optimization of noise removal feature extraction methods.

In the optimization of feature fusion, compared to some methods exploring multiangle fusion of bitemporal features like SST-Former and SATNet, our simple addition and subtraction for temporal-related feature fusion already outperform some of the latest methods. Experimental results further validate the effectiveness of our proposed methods and modules.

2) Visual Comparison: In terms of visual comparison, we observe that FDIEG-UNet achieves better visual results compared to other models, generating grayscale images with fewer noisy points.

However, compared to other DL algorithms, FDIEG-UNet does not perform well in detecting complex situations that mix invariant and changing regions, especially in CD with edges and complex pixels. Our analysis suggests that the reason may be due to our excessive focus on semantic information, resulting in the loss of some detailed information in the original pixel space.

3) Computational Complexity: As shown in Table II, we achieved better change detection performance compared with other methods with similar or lower computational complexity. Table III displays the computational complexity analysis of the important parts of our proposed method. H, W, and D represent the height, width, and number of channels of the feature map, respectively. k is the kernel size of the convolution operation. Due to the complexity of the fast Fourier transform (FFT) algorithm being $O(L \log L)$, the efficiency (HWD) of feature learning in FDGJLS is much higher compared to transformer and 3-D convolutional networks.

D. Ablation Studies

1) Training Sample Sizes: We compared and analyzed different training sample sizes to see if we still have advantages under different data sizes (2%, 5%, 10%). As shown in Table II, here we choose some methods with better results for comparison. For the Farmland dataset, the proposed method performs less effectively than STSS-Net and DCENet in the cases of 5%–10%. This may be due to the fact that the Farmland dataset itself is easy to distinguish, and the imaging is rarely influenced by external factors. Changes can be easily identified by methods optimized for spectral and spatial feature extraction. Although our method is effective, its performance improvement is not as good as STSS-Net. For other datasets, we achieve the best results with different training sample sizes. This indicates that our method performs well not only with small samples but also with large

TABLE IV
ABLATION STUDY OF OUR FDGJLS (F) MODULE, ULDT (U) MODULE, AND SSTDFE (S) MODULE ON THREE HSI-CD DATASETS. THE DATA IS KC

F	U	S	Farmland	Santa Barbara	Bay Area
✓	✓	✓	**92.28**	**97.50**	**97.10**
✓	×	✓	91.50	96.51	96.08
✓	✓	×	91.70	96.65	96.19
✓	×	×	90.82	95.88	95.82

TABLE V
ABLATION STUDY OF SSTDFE MODULE ON FARMLAND DATASET. 0 REPRESENTS UNCHANGED SAMPLES AND 1 REPRESENTS CHANGED SAMPLES

SSTDFE	iou		F1	
	0	1	0	1
✓	95.00	**90.55**	97.43	**94.57**
×	94.37	87.74	97.10	93.47

TABLE VI
ABLATION STUDY OF SSTDFE MODULE ON SANTA BARBARA DATASET. 0 REPRESENTS UNCHANGED SAMPLES AND 1 REPRESENTS CHANGED SAMPLES

SSTDFE	iou		F1	
	0	1	0	1
✓	98.09	**96.98**	99.03	**98.46**
×	96.81	95.12	98.38	97.50

TABLE VII
ABLATION STUDY OF SSTDFE MODULE ON BAY AREA DATASET. 0 REPRESENTS UNCHANGED SAMPLES AND 1 REPRESENTS CHANGED SAMPLES

SSTDFE	iou		F1	
	0	1	0	1
✓	96.94	**97.33**	98.44	**98.65**
×	95.68	95.16	97.79	97.52

TABLE VIII
ABLATION STUDY OF FDGJLS MODULE ON THREE DATASET. THE DATA IS KC

+3D-FFT	Farmland	Santa Barbara	Bay Area
FDGJLS	**92.28**	**97.59**	**96.80**
S-TE	91.63	96.92	96.08
3D-CNN	91.81	97.03	96.19

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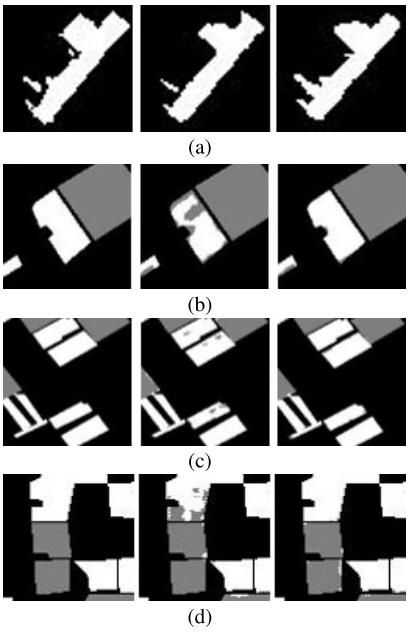


Fig. 10. Display the effectiveness of enhancing positive sample recognition (ground truth, ablation on SSTDFE, SSTDFE from left to right).

samples. In fact, the proposed method's performance with small samples training is better than other methods trained with large samples.

2) Ablation on ULDT: We perform ablation by removing the ULDT module from the model. As shown in Table IV, in the comparison of the first and second experiments, as well as the comparison of the third and fourth experiments, our kappa score continued to decrease after removing this module. This indicates that in extracting temporal-related fusion features $A+B+2U$, the feature domain was effectively optimized by removing the sum of bitemporal invalid style information, improving the saliency of the features.

3) Ablation on SSTDFE: We perform ablation by removing the SSTDF module from the model. As shown in Table IV, in the comparison of the first and third experiments, as well as the comparison of the second and fourth experiments, our kappa score continued to decrease after removing this module. This indicates that effective optimization of features in the feature domain has been achieved in extracting time-related differential features $A-B$, improving the recognition ability of changed samples.

As shown in Tables V–VII, we compared the scores of changing and unchanged samples using three indicators on three datasets. We can see that the addition of a new feature fusion method in the SSTDFE module greatly improves the score of changed samples. As shown in Fig. 10, we can demonstrate this by zooming in on the visualized prediction results. This provides a powerful explanation for the improvement of the equality of feature description between changed and unchanged samples in CD.

4) Ablation on FDGJLS: Because the FDGJLS module is the foundation for providing information for optimizing feature domains for subsequent modules, we use different methods to replace the FDGJLS module in the model for ablation. As shown in Table VIII, we will replace the 3-D FFT-based global joint feature learning method with transformer-based spatial (S-TE) global feature learning and 3-D CNN-based joint feature learning, respectively. It is clear to see the superiority of the FDGJLS module in feature extraction and the improvement of HSI-CD performance.

E. Effect of Parameters

a) Loss coefficient: We have two loss functions. To find the optimal weight sum of the two loss functions, we tested five different weights $\lambda = \{0.1, 0.2, 0.5, 1.0, 1.2\}$. As shown in Fig. 11(a), we found that the best performance was achieved when the weight was 1.0.

b) Training samples: To investigate the effect of training samples on experimental results, we conducted a series of experiments using different sample proportions of 0.1%, 1%, 2%, 10%, and 20% on three datasets. As shown in Fig. 11(b), the performance of CD significantly improves with the increase in dataset size.

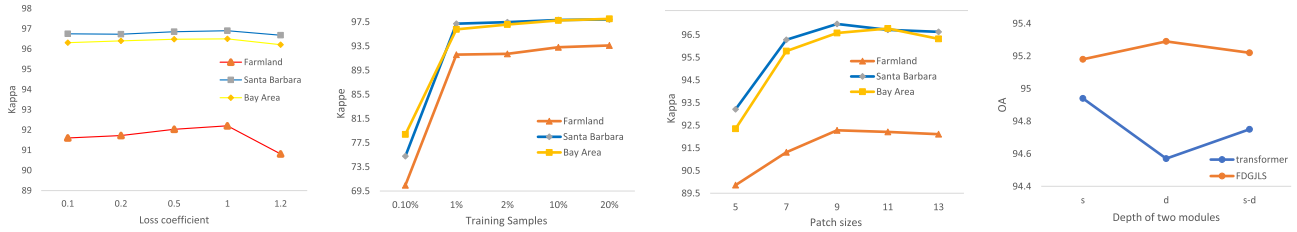
c) Patch sizes: Our model is a multiscale method, but HSI-CD is a patch-based task. We explored patch sizes on three datasets. To adapt our model, we removed the deep encoder when the patch size was too small. As shown in Fig. 11(c), we can see that our CD performs best when the patch size is 9.

d) Depth of module: The sequential transformer and FDGJLS module with global–joint learning exhibit different performances in shallow and deep feature learning, and the layer placement is an important parameter. We only keep the baseline and conduct experiments on the Farmland dataset based on it. To find the optimal layer for both, we conducted tests on a shallow-to-deep hierarchy $t = s-d, s, d$, where $s-d$ indicates that the module is added to both shallow and deep encoders, s indicates that it is added only to the last two layers of the encoder, and d indicates that it is added only to the first two layers of the encoder. As shown in Fig. 11(d), we compared the OA scores of depth of module in the Farmland dataset. We added the sequential transformer to the first two layers and the FDGJLS module with global–joint learning to the last two layers.

V. Conclusion

In this article, we design a network called FDIEG-UNet based on a proposed dual-temporal information comparison composition approach. In FDIEG-UNet, we use the FDGJLS module for global and joint feature learning and information separation as an information provider. Then, we use the SSTDFE module and the ULDT module for frequency-domain processing to optimize the information in the feature domain. It is worth noting that we not only achieve state-of-the-art performance with a small training dataset, but also improve the classification performance of changed samples. However, our proposed method has certain drawbacks. We focus too much on semantic information, which results in the loss of some spatial details and the inability to restore to the pixel space. In the visualization of experiments, we can see that the detection effect on edges and subtle changes is not good. In our future work, we will further address these issues by incorporating fine-grained information in the original pixel space.

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(a) (b) (c) (d)

Fig. 11. Scores of FDIEG-UNet under different parameters in three datasets. (a) Loss coefficient. (b) Training samples. (c) Patch sizes. (d) Depth of two modules in Farmland dataset.

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